

SOFTWARE PROJECT PROPOSAL

Legacy Scale — Agency Education Platform

Premium Video LMS · Claude AI Toolkit · Gamified Community · Live Events

Prepared for Keanu Vasquez · legacyscale.co

Billing Model: Hourly — Weekly Accumulated Hours

Rate: \$16 USD / Hour

Total NTE: \$12,000 (750 hours)

Timeline: 22 Weeks Standard · 15 Weeks with Claude Code

CONFIDENTIAL — Prepared for Client Review

1. Executive Summary

1.1 What We Are Building

This proposal covers the complete development of a premium single-tenant SaaS platform for Legacy Scale — Keanu Vasquez's 7-Figure Agency education brand. The platform unifies four core products under one subscription: a Claude AI-powered coaching toolkit, a structured HD video LMS, a gamified community portal, and a live event calendar. Every member logs in to one platform and accesses their own data, their own progress, and their own AI history.

1.2 What Makes This Different from Skool or Kajabi

- Built specifically for agency education — not a generic platform Keanu has to work around
- Claude AI tools built in: Marketing Copy Generator, Agency Proposal Builder, AI Business Coaching Agent
- Gamification creates daily retention loops — members compete, track streaks, and earn badges
- Full ownership of data, branding, and features — no platform risk, no revenue share
- Single subscription replaces multiple tools members would otherwise pay for separately

1.3 Architecture: Single-Tenant SaaS

This is a single-tenant SaaS platform. One codebase, one database, one brand. Every user logs in and sees only their own data, scoped by `user_id`. No multi-tenancy, no subdomain routing, no per-tenant branding. This keeps the architecture clean and the timeline realistic.

SINGLE-TENANT MEANS: One platform. One brand. Many paying members. Every member sees only their own courses, progress, AI history, and points. Think Skool — but built for Keanu's brand with AI tools built in.

1.4 Billing at a Glance

Hourly at \$16 USD per hour, billed weekly with an itemized timesheet. Each of the seven phases has a Not-to-Exceed (NTE) cap. You know your maximum cost before each phase starts.

2. Platform Analysis

2.1 What Legacy Scale Offers

Based on legacyscale.co, Keanu Vasquez teaches Marketing Fulfillment Arbitrage. The platform delivers:

- 7-Figure Agency Program: the core training curriculum
- High Ticket Marketing Masterclass
- Marketing Bootcamp
- Millionaire Mentorship Days 1–30
- Private community group access

2.2 Competitive Comparison

Platform	AI Tools	Video LMS	Gamification	Live Events	Gap
Teachable	—	✓	—	—	No AI, no community, no gamification
Skool	—	✓	✓	✓	No AI tools, generic, not agency-focused
Kajabi	—	✓	—	✓	No AI, no gamification, expensive, generic
Whop	—	—	—	—	Marketplace only, not a training platform
Legacy Scale Platform	✓	✓	✓	✓	Built specifically for Keanu's brand and curriculum

3. Technology Stack

Every technology is production-proven and selected for maintainability, developer velocity, and long-term scalability.

3.1 Frontend

Layer	Technology	Rationale
Framework	Next.js 14	Vercel/Linear aesthetic; SSR, image optimization, clean routing
Language	TypeScript	Eliminates runtime bugs; essential for large maintainable codebase
Styling	Tailwind CSS 3	Pixel-perfect premium UI; dark/light mode
Components	ShadCN/UI	Fully customizable Tailwind-native; premium dark/light mode
State	Zustand	Lightweight TypeScript-native global state management
Data Fetching	TanStack Query	Caching, background refetch, pagination automatically
Forms	React Hook Form + Zod	Shared validation schema between frontend and backend
Charts	Recharts + Tremor	Custom viz + BI dashboard components
Video	Mux Player React	Adaptive streaming and DRM automatically
Calendar	FullCalendar React	Google Calendar adapter built in
Animations	Framer Motion	Micro-interactions, leaderboard animations, premium UX
Auth	Clerk embedded	OAuth, MFA, social login, JWT — zero custom auth code

3.2 Backend

Layer	Technology	Rationale
Runtime	Node.js 20 LTS	Handles AI streaming and WebSocket connections efficiently
Framework	NestJS	TypeScript-native modular; scales cleanly from day one
ORM	Prisma	Schema-first, type-safe; user_id middleware enforcement

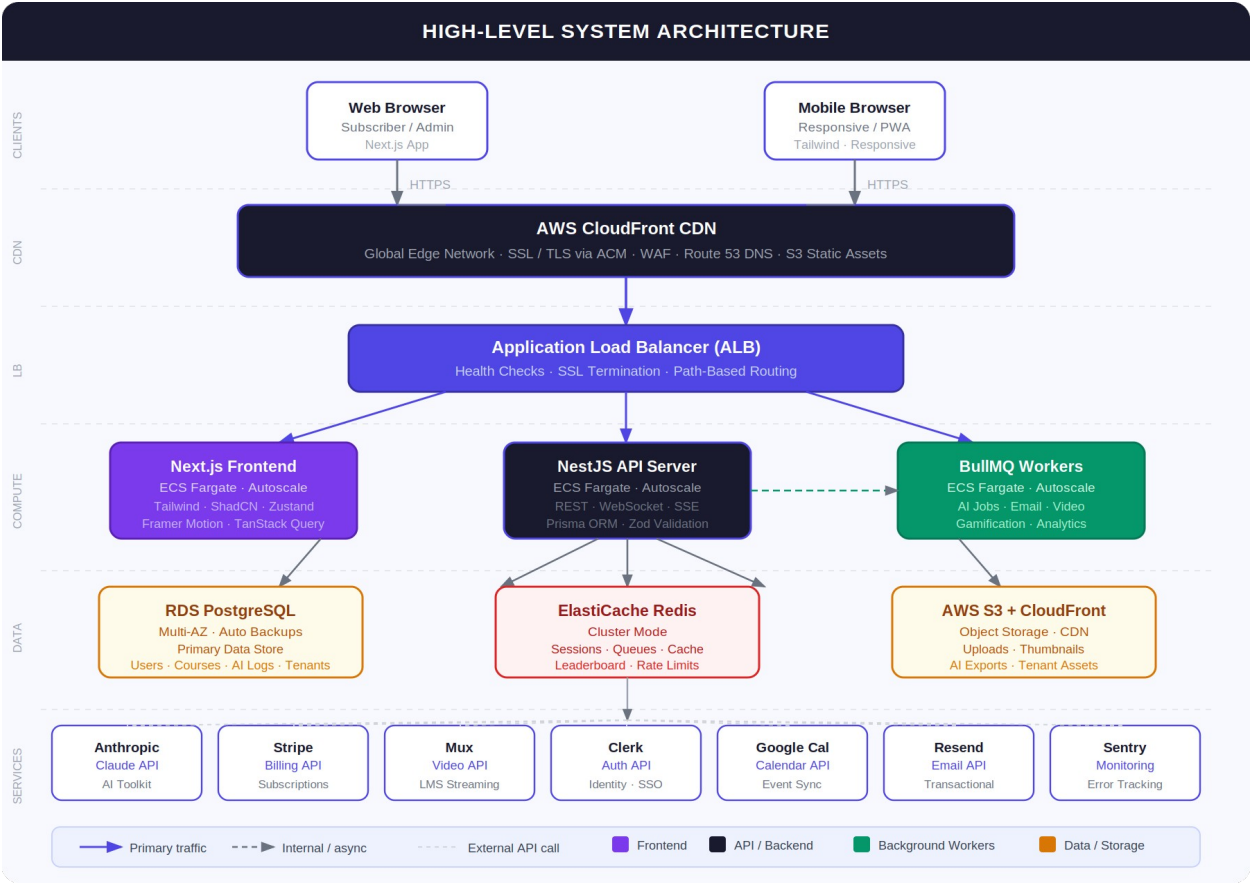
Layer	Technology	Rationale
Queues	BullMQ + Redis	Background jobs: AI, email, video webhooks, gamification
Validation	Zod (shared)	One schema on both frontend and backend
Email	Resend + React Email	Modern email API with React templates
Uploads	AWS S3 presigned URLs	Direct browser-to-S3; API never handles binary data
Auth	Clerk backend SDK	JWT verification; ADMIN and MEMBER roles only
Payments	Stripe Node SDK	Recurring subscriptions, webhooks, billing portal

3.3 AI Stack (Claude API)

Component	Choice	Notes
Primary LLM	Anthropic Claude (claude-sonnet-4-20250514)	As specified; best for structured output and instruction-following
Streaming	Server-Sent Events (SSE)	Real-time token streaming to the UI
Prompt Registry	Admin-editable DB templates	Prompts updated by Keanu without code deployments
Coaching Agent	Redis conversation store	Multi-turn context maintained per user session
Cost Controls	Per-user token logging + daily limits	Usage tracked per feature; enforced by subscription status

4. System Architecture

4.1 High-Level Architecture Diagram



4.2 Single-Tenant Data Flow

SINGLE-TENANT REQUEST FLOW User logs in via Clerk (email/password, Google OAuth, magic link) | v JWT contains: `userId`, `role` (ADMIN or MEMBER), `subscriptionStatus` | v NestJS AuthGuard verifies JWT on every request | v All queries scoped: `WHERE user_id = :userId` | v User sees only their own data — no cross-user data access possible

4.3 AI Request Lifecycle

AI REQUEST LIFECYCLE (Marketing Copy Generator) Member submits: product/service, target client, platform, goal | v `POST /api/ai/marketing-copy` — Zod schema validation | v Rate limit check: has this user hit their daily limit? | v Subscription check: is this user's subscription active? | v Prompt template loaded from admin-editable registry | v SSE streams Claude tokens to browser in real time | v On complete: log usage (tokens, cost, `user_id`) to `ai_usage_logs`

5. Feature Breakdown by Module

MODULE 1: Auth & User Management

Property	Detail
Purpose	Secure identity and role-based access. Two roles: ADMIN (Keanu) and MEMBER (subscribers).
Frontend	Login, register, forgot password, profile settings, avatar upload, role-based UI
Backend	Clerk integration, user sync webhooks, JWT guard, ADMIN/MEMBER roles, profile CRUD
Database	users
APIs	Clerk API, S3
Complexity	Medium
Est. Hours	40–55 hrs

MODULE 2: Subscription & Billing

Property	Detail
Purpose	Members pay via Stripe. Access gated by subscription status.
Frontend	Pricing page, Stripe Checkout, billing portal, upgrade prompts, payment status pages
Backend	Stripe Checkout, subscription webhooks, tier enforcement middleware, billing portal
Database	subscriptions, payment_events
APIs	Stripe API
Complexity	High
Est. Hours	45–58 hrs

MODULE 3: Video LMS

Property	Detail
Purpose	HD video training: 7-Figure Agency, Bootcamp, Masterclass, Mentorship.
Frontend	Course library, Mux Player, progress bars, module nav, completion UI, certificates, search

Property	Detail
Backend	Course/module/lesson CRUD, Mux upload + webhook, progress API (30s), enrollment, certs
Database	courses, modules, lessons, enrollments, lesson_progress, certificates
APIs	Mux Video API
Complexity	High
Est. Hours	80–105 hrs

MODULE 4: Event Calendar

Property	Detail
Purpose	Live training, coaching, and mentorship scheduling with Google Calendar sync.
Frontend	FullCalendar views, event creation, RSVP, Google Cal connect, iCal export
Backend	Event CRUD, Google Calendar OAuth (bidirectional), BullMQ token refresh, RSVP, reminders
Database	events, event_rsmps, calendar_integrations
APIs	Google Calendar API
Complexity	High
Est. Hours	58–75 hrs

MODULE 5: Gamification Engine

Property	Detail
Purpose	Points, streaks, badges, and real-time leaderboard for daily engagement.
Frontend	Live leaderboard (WebSocket), XP display, streak counter, badges, activity feed, animations
Backend	Event-driven points engine, streak logic, badge awards, Redis ZADD, NestJS WebSocket gateway
Database	user_points, point_events, badges, user_badges, streaks
APIs	Internal WebSocket
Complexity	High
Est. Hours	62–80 hrs

MODULE 6: Claude AI Toolkit

Property	Detail
Purpose	Three Claude tools for members: Marketing Copy, Proposal Builder, Coaching Agent.
Frontend	AI tool hub, streaming output UI, AI history, token meter, coaching chat
Backend	Prompt registry (admin-editable), Claude SSE integration, rate limiting, usage logging, Redis sessions
Database	ai_prompts, ai_sessions, ai_messages, ai_usage_logs
APIs	Anthropic Claude API
Complexity	Very High
Est. Hours	82–108 hrs

MODULE 7: Dashboard

Property	Detail
Purpose	Personalized member home: progress, events, leaderboard rank, AI usage.
Frontend	Welcome widget, course progress, upcoming events, leaderboard rank, quick-launch tools
Backend	Dashboard aggregation API; notification service
Database	Aggregates from all modules
APIs	Internal
Complexity	Medium
Est. Hours	32–44 hrs

MODULE 8: Admin Panel

Property	Detail
Purpose	Full platform management for Keanu.
Frontend	User management, course management, AI prompt editor, subscription overview, BI analytics
Backend	Admin-only guards, bulk operations, metrics aggregation, audit logging
Database	admin_actions
APIs	Stripe, Mux

Property	Detail
Complexity	High
Est. Hours	52–66 hrs

MODULE 9: Notifications

Property	Detail
Purpose	In-app and email alerts for all platform events.
Frontend	Notification bell, notification panel, preference settings
Backend	WebSocket push + Resend email, BullMQ email queue
Database	notifications, notification_preferences
APIs	Resend
Complexity	Medium
Est. Hours	28–40 hrs

MODULE 10: Analytics & Reporting

Property	Detail
Purpose	BI dashboard for Keanu; personal analytics for each member.
Frontend	Admin BI: DAU/MAU, revenue, completions, attendance (Recharts/Tremor); member: personal progress
Backend	PostgreSQL aggregation + materialized views, CSV export
Database	analytics_events (partitioned monthly)
APIs	Internal
Complexity	High
Est. Hours	44–60 hrs

6. Database Design

6.1 Single-Tenant Principle

Single-tenant PostgreSQL database. No tenant tables, no tenant_id columns. Every user's data is scoped by user_id via Prisma middleware. The schema is exactly as complex as it needs to be — nothing more.

DATA ISOLATION: Every query touching user data includes WHERE user_id = :userId Enforced by Prisma middleware — impossible to access another user's data.

6.2 Entity Relationships

users (1) —< subscriptions users (1) —< enrollments —> courses courses (1) —< modules —< lessons
users (1) —< lesson_progress —> lessons users (1) —< user_points users (1) —< user_badges —>
badges users (1) —< ai_sessions —< ai_messages users (1) —< ai_usage_logs events (1) —<
event_rsvps —> users users (1) —< notifications users (1) —< audit_logs

6.3 Core Tables

Table	Key Columns	Notes
users	id, clerk_id, email, role (ADMIN/MEMBER), created_at	Two roles only; no tenant complexity
subscriptions	id, user_id, stripe_customer_id, stripe_sub_id, status, period_end	status: active/past_due/canceled/trialing
courses	id, title, slug, thumbnail_url, status, order_index	DRAFT/PUBLISHED/ARCHIVED
modules	id, course_id, title, order_index	Sections within a course
lessons	id, module_id, title, mux_asset_id, mux_playback_id, duration_seconds, order_index, status	Mux IDs populated after webhook
enrollments	id, user_id, course_id, enrolled_at, completed_at	completed_at null until all lessons done
lesson_progress	id, user_id, lesson_id, watch_seconds, completed, completed_at	UNIQUE (user_id, lesson_id); updated every 30s
events	id, title, start_at, end_at (UTC), google_event_id, max_rsvps	All times in UTC
event_rsvps	id, user_id, event_id, rsvp_at	UNIQUE (user_id, event_id)
calendar_integrations	id, user_id, google_access_token, refresh_token, expires_at	Refreshed by BullMQ job

Table	Key Columns	Notes
user_points	id, user_id, points, source_type, source_id, created_at	Append-only audit trail
streaks	id, user_id, current_streak, longest_streak, last_activity_date	Updated on qualifying actions
badges + user_badges	id, name, icon_url, trigger_type, trigger_value user_id, badge_id, awarded_at	UNIQUE (user_id, badge_id)
ai_prompts	id, feature_key, system_prompt, user_prompt_template, model, version	Admin-editable; versioned
ai_sessions + ai_messages	id, user_id, feature_key session_id, role, content, tokens_used	Append-only conversation history
ai_usage_logs	id, user_id, feature_key, input_tokens, output_tokens, cost_usd, created_at	Cost tracking + rate limiting
notifications	id, user_id, type, title, body, read, action_url, created_at	In-app store
audit_logs	id, user_id, action, entity_type, entity_id, metadata, ip, created_at	Immutable
analytics_events	id, user_id, event_type, properties (JSONB), created_at	Partitioned by month

7. AWS Infrastructure

7.1 Environments

Environment	Purpose	Infrastructure	Trigger
local	Development	Docker Compose (Postgres, Redis, Next.js, NestJS)	Manual
staging	QA + preview	Single ECS task per service, RDS t3.micro, Redis t3.micro	Push to main
production	Live platform	Multi-AZ ECS, RDS Multi-AZ, Redis cluster	Manual approval

7.2 Estimated Monthly AWS Costs

Service	MVP/mo	Growth/mo	Notes
ECS Fargate	~\$45	~\$180	3 services, 2 tasks min each
RDS PostgreSQL (t3.medium)	~\$55	~\$200	Multi-AZ in production
ElastiCache Redis	~\$18	~\$70	Single node MVP; cluster at growth
CloudFront + S3	~\$10	~\$40	Assets, thumbnails, exports
ALB (x2) + NAT Gateway	~\$55	~\$110	Load balancers + outbound traffic
Route 53 + ACM + Secrets + Monitoring	~\$16	~\$37	DNS, SSL, secrets, logs
TOTAL AWS	~\$199/mo	~\$637/mo	Excludes Mux, Clerk, Stripe, Claude API

8. Development Timeline

8.1 Phase Overview

Phase	Weeks	Deliverable	NTE Hours	NTE Cost
Phase 0: Foundation	1–2	AWS, CI/CD, design system	70 hrs	\$1,120
Phase 1: Auth & Billing	3–5	Clerk auth + Stripe subscriptions	100 hrs	\$1,600
Phase 2: Video LMS	6–9	Full LMS with progress + certificates	120 hrs	\$1,920
Phase 3: Claude AI Toolkit	10–13	3 AI tools with streaming	120 hrs	\$1,920
Phase 4: Calendar + Gamification	14–16	Google Cal + live leaderboard	110 hrs	\$1,760
Phase 5: Admin + Analytics	17–18	Admin panel + BI dashboard	100 hrs	\$1,600
Phase 6: QA + Security + Launch	19–22	Testing, security, production deploy	130 hrs	\$2,080
TOTAL	22 weeks	Full platform	750 hrs	\$12,000

8.2 With Claude Code: 15 Weeks

Claude Code as AI pair programmer compresses the timeline by 7 weeks at no extra cost to the client. Same \$16/hr. Same quality. Same scope. Delivered faster.

Phase	Standard	With Claude Code	Time Saved	How Claude Code Helps
Phase 0	2 wks	1 wk	-1 wk	Boilerplate, configs, Docker, CI/CD generated instantly
Phase 1	3 wks	2 wks	-1 wk	CRUD, Stripe webhooks, Prisma schema drafted by Claude Code
Phase 2	4 wks	3 wks	-1 wk	Mux integration, progress API boilerplate scaffolded

Phase	Standard	With Claude Code	Time Saved	How Claude Code Helps
Phase 3	4 wks	3 wks	-1 wk	Claude Code writes its own integration (meta but fast)
Phase 4	3 wks	2 wks	-1 wk	Points engine, leaderboard, BullMQ jobs scaffolded
Phase 5	2 wks	2 wks	same	Admin tables and charts generated quickly
Phase 6	4 wks	2 wks	-2 wks	Unit tests, E2E scaffolding, security checklist, deploy configs
TOTAL	22 wks	15 wks	-7 wks	

9. Project Gantt Chart

Standard 22-week plan. Each block (■) = one week.

Phase	Window	Focus	Weeks
			1—————22
P0 — Foundation	W1–W2	AWS, design system	■ ■
P1 — Auth & Billing	W3–W5	Clerk, Stripe	■ ■ ■
P2 — Video LMS	W6–W9	Mux, courses, certs	■ ■ ■ ■
P3 — AI Toolkit	W10–W13	3 Claude tools	■ ■ ■ ■
P4 — Calendar + Gamification	W14–W16	Google Cal, leaderboard	■ ■ ■
P5 — Admin + Analytics	W17–W18	Admin panel, BI	■ ■
P6 — QA + Launch	W19–W22	Testing, security, deploy	■ ■ ■ ■

MILESTONES: M1 W5 Auth+Billing | M2 W9 LMS | M3 W13 AI tools | M4 W16 Calendar+Gamification | M5 W18 Admin | M6 W22 Launch

10. Hour Estimation

Module / Phase	Conservative	Realistic	Maximum
Phase 0: Foundation	55	70	85
Auth & User Management	38	50	62
Subscriptions & Billing	40	55	68
Video LMS	75	100	128
Event Calendar (Google Cal)	55	70	88
Gamification Engine	58	75	95
Claude AI Toolkit (3 tools)	78	102	130
Dashboard	30	40	52
Admin Panel	48	62	78
Notifications	25	36	48
Analytics & BI Dashboard	40	56	72
QA + Testing	52	72	98
Security Hardening	22	32	44
DevOps + CI/CD	32	48	62
Buffer	48	72	110
TOTAL	696	940	1,220

NTE CAPS are set at the realistic estimate with buffer. With Claude Code, expect to finish 20-30% under the caps — you pay only actual hours.

11. Billing Model & Payment Plan

11.1 Rate

Hourly Rate: \$16 USD per hour | Billing: Weekly accumulated hours | Invoice: Itemized timesheet every Friday

This rate represents senior-level full-stack engineering: AI integration (Claude API), SaaS architecture, AWS infrastructure, and DevOps. Philippine-based talent delivering approximately 79-82% cost savings vs equivalent US/AU developer rates (\$75-\$95/hr).

11.2 Phase-Level NTE Caps

Phase	Scope	NTE Hours	Max Cost	When Billed
Phase 0	Foundation, AWS, Design System	70 hrs	\$1,120	Weekly
Phase 1	Auth + Billing	100 hrs	\$1,600	Weekly
Phase 2	Video LMS	120 hrs	\$1,920	Weekly
Phase 3	Claude AI Toolkit	120 hrs	\$1,920	Weekly
Phase 4	Calendar + Gamification	110 hrs	\$1,760	Weekly
Phase 5	Admin Panel + Analytics	100 hrs	\$1,600	Weekly
Phase 6	QA + Security + Launch	130 hrs	\$2,080	Weekly
TOTAL NTE	Full Platform	750 hrs	\$12,000	Weekly throughout

COST COMPARISON: US/AU developer (same scope, 750 hrs x \$75-\$95/hr) = \$56,250-\$71,250. This engagement: \$12,000 maximum. Saving: \$44,000-\$59,000.

11.3 Weekly Billing Process

- Hours logged daily with task-level descriptions
- Timesheet submitted every Friday with accumulated hours
- Invoice issued same day, itemized by task and module
- Payment due within 5 business days
- \$640 kickoff deposit applied against the first week of billing

11.4 Kickoff Deposit

A deposit of \$640 USD (40 hours at \$16/hr) is required before work begins. This is standard practice across all professional services — lawyers require a retainer, designers require a

deposit, contractors require materials money upfront. It protects both parties: it confirms Keanu's commitment before significant development time is invested, and it ensures the first week of foundational work (AWS setup, CI/CD, project structure) is covered before the first weekly invoice is issued.

The deposit is applied directly against the first week of billing — it is not an additional charge. If preferred, this can be held in Upwork escrow, where neither party controls the funds until the first milestone is approved.

11.5 Scope Change Policy

Any requirement not in this proposal is a change request. Scoped, estimated in writing, approved before work begins. Billed at \$16/hr. Does not affect existing phase NTE caps.

12. Risk Assessment

Risk	Likelihood	Impact	Mitigation
Scope creep	High	High	Written change request process; approval before any out-of-scope work begins
Claude API cost overrun	Medium	High	Per-user daily token limits; usage logged per feature; admin cost dashboard
Google Calendar OAuth	Medium	Medium	BullMQ refresh job 30 min before expiry; graceful re-auth UI flow
Stripe webhook idempotency	Low	High	Idempotency keys + event deduplication in DB
Mux video webhook failure	Low	High	Retry logic with backoff; polling fallback; admin alert
Credential exposure	Low	Very High	AWS Secrets Manager; no hardcoded credentials; npm audit
Client feedback delays	Medium	Medium	Weekly deliverables; 3-day review window per phase

13. Security Architecture

13.1 Authentication

- Clerk: password hashing, sessions, OAuth 2.0, brute-force protection, MFA
- JWTs short-lived (1 hr); Clerk manages refresh automatically
- Two roles only: ADMIN (full access) and MEMBER (own data only)

13.2 Data Security

- All user-specific queries require WHERE user_id = :userId via Prisma middleware
- Member cannot access another member's data — impossible at the query layer
- All data encrypted at rest (RDS) and in transit (TLS 1.2+ via CloudFront)

13.3 API Security

- All routes protected by NestJS AuthGuard by default
- Admin routes reject MEMBER role instantly at the guard level
- All input validated by Zod; malformed requests rejected before business logic
- Rate limiting: global 100 req/min per IP + per-user AI daily limits
- Helmet.js: CSP, HSTS, X-Frame-Options on all responses

13.4 Infrastructure

- VPC: all ECS tasks + databases in private subnets; only ALB publicly accessible
- AWS Secrets Manager: all API keys injected at container runtime; zero hardcoded credentials
- RDS: daily automated backups, 7-day retention, point-in-time recovery enabled

14. Deployment & CI/CD

PIPELINE On PR to staging: ESLint + TypeScript + Jest + build verification On merge to staging: Docker build + push ECR + rolling ECS deploy On manual approval to production: RDS snapshot + Prisma migrations + ECS rolling deploy

Rollback & Monitoring

- ECS retains previous task definition; rollback in under 3 minutes
- All DB migrations backwards-compatible by design
- Sentry: frontend + backend error tracking
- CloudWatch: ECS health, RDS performance, custom alarms
- External uptime monitoring (Uptime Robot): SMS/email on downtime

15. Post-Launch Maintenance

15.1 Immediate Post-Launch

- Daily monitoring of Sentry, CloudWatch, and Stripe dashboard
- Critical bugs: 24hr resolution; major bugs: 72hr
- Claude API cost monitoring vs projections

15.2 Optional Retainer Plans

Plan	Monthly Rate	Coverage
Basic	\$800–\$1,200/mo	Monitoring, critical bug fixes, monthly dependency updates
Standard	\$2,000–\$3,000/mo	Basic + up to 20 hrs/month feature development
Premium	\$3,500–\$5,000/mo	Standard + priority SLA + up to 40 hrs/month

16. Final Recommendation

16.1 Build Sequence

- 1. Auth + Billing — Everything requires paying members to exist first
- 2. Video LMS — First value delivery to subscribers
- 3. Claude AI Toolkit — The wow moment that differentiates Legacy Scale from every generic Skool community
- 4. Event Calendar — Live sessions prove the community is active
- 5. Gamification — Compounding daily retention once content and tools exist
- 6. Admin + Analytics — Can trail member-facing features by 1-2 weeks

16.2 Investment Summary

Item	Detail
Hourly Rate	\$16 USD / hour
Billing Cycle	Weekly, accumulated hours with itemized timesheet
Kickoff Deposit	\$640 USD (applied against the first week of billing)
Total NTE	\$12,000 USD (750 hours x \$16/hr)
Standard Timeline	22 weeks from kickoff to production launch
With Claude Code	15 weeks from kickoff to production launch
Scope Change Policy	Written approval required; \$16/hr; does not affect phase NTE caps
Post-Launch Support	Optional retainer plans from \$800/mo

TOTAL SAVINGS: This engagement delivers the equivalent of \$56,250-\$71,250 of senior engineering at a maximum of \$12,000 — a saving of \$44,000-\$59,000 vs US/AU rates.

END OF PROPOSAL